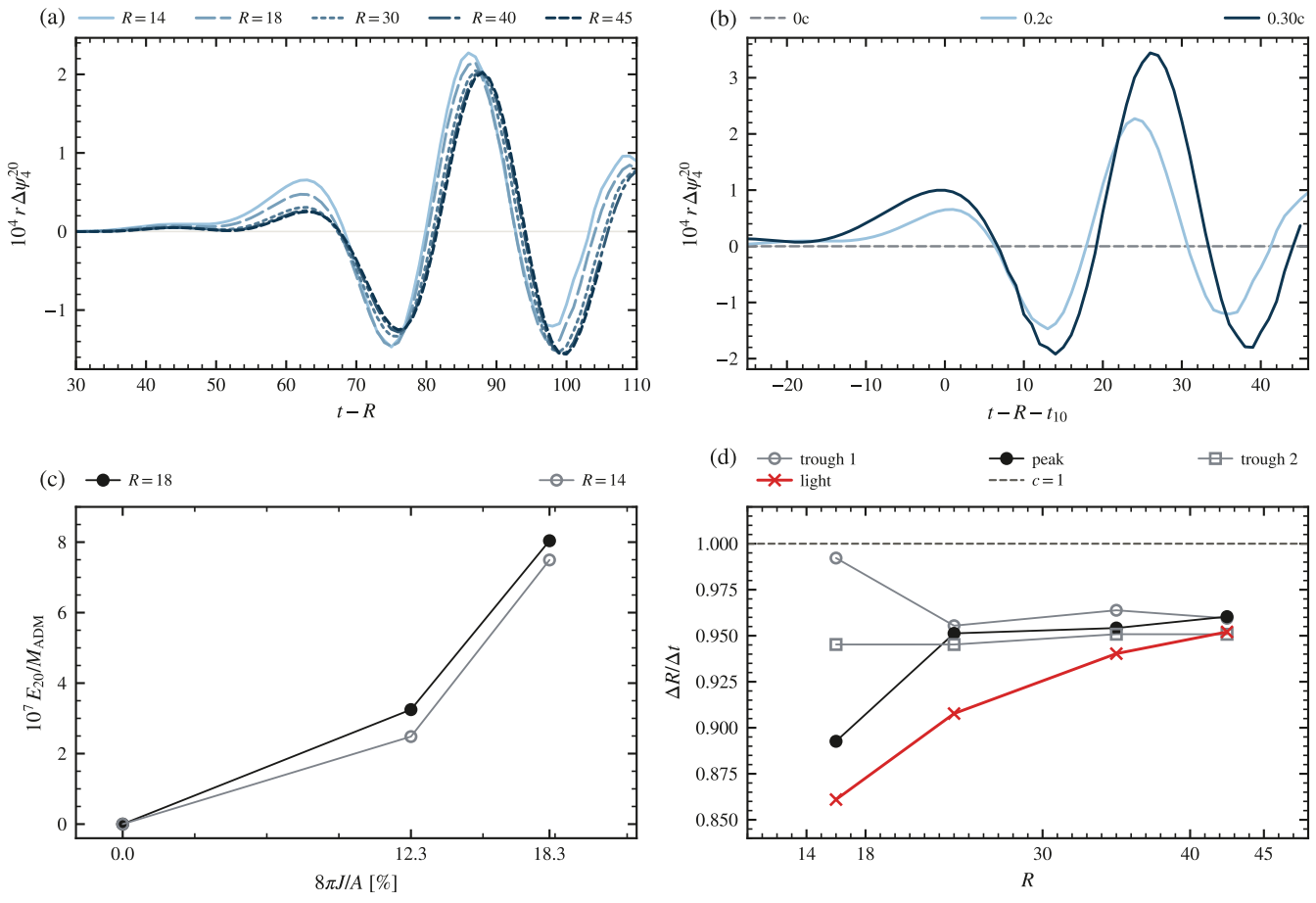
(a)
R = 14
R = 18
R = 30
R = 40
R = 45
$10^4\, r\, \Delta\psi_4^{20}$
$t - R$
(b)
0c
0.2c
0.30c
$10^4\, r\, \Delta\psi_4^{20}$
$t - R - t_{10}$
(c)
R = 18
R = 14
$10^7\, E_{20}/M_{\rm ADM}$
$8\pi J/A$ [%]
0.0
12.3
18.3
(d)
trough 1
light
peak
c = 1
trough 2
$\Delta R/\Delta t$
R